Name

Mathematics Stage 5 Path

Further Algebraic Fractions

Book 1 Simplifying algebraic fractions using factorisation

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Syllabus Content

MA5-ALG-P-02 selects and applies appropriate algebraic techniques to operate with algebraic fractions, and expands, factorises and simplifies algebraic expressions

- Simplify algebraic expressions involving algebraic fractions using factorisation

Simplifying Algebraic Fractions by Factorisation

Review

☐ Factorise

a $6x + 9$	**b** $2x^2 + 4x$	**c** $x^2 - 25$

☐ Simplify fractions

a $\frac{4x}{6x}$	**b** $\frac{3x+9}{18}$	**c** $\frac{4x+8}{2}$

Identify whether a fraction can be simplified

A fraction $\frac{A}{B}$ can be simplified if and only if A and B share a common factor other than 1.

✗	✓	✓	✓	✗
$\frac{2x+3}{3}$	$\frac{4x+8}{2}$	$\frac{(x-3)(x+3)}{x+3}$	$\frac{(x+2)(x+3)}{x+2}$	$\frac{(x+2)(x+3)}{x-2}$

Decide whether these fractions can be simplified.

a $\frac{3x+9}{3}$	**b** $\frac{x+5}{x+10}$	**c** $\frac{(x-4)(x+4)}{x+4}$
d $\frac{x^2+3x}{x}$	**e** $\frac{(x+1)(x+2)}{x+5}$	**f** $\frac{(x+4)(x+2)}{x+2}$

Simplifying Algebraic Fractions by Simple Factorisation

1. Factorise the numerator and denominator.
2. Cancel any common factors.

Example Simplify algebraic fractions by simple factorisation

$\frac{6x+8}{6}$ 1. $\frac{2(3x+4)}{6}$ 2. $\frac{3x+4}{3}$	We factorise to see if the numerator and denominator have common …… $\frac{3x+4}{3}$ can't be simplified further because the numerator and denominator do not share any common ……

Guided Practice

Simplify

a $\frac{5a+10}{5}$

$= \frac{5(\quad)}{5}$

$=$

$a+2$

b $\frac{8}{4d-2}$

$= \frac{8}{2(\quad)}$

$=$

$\frac{4}{2d-1}$

c $\frac{2p+2q}{p+q}$

2

d $\frac{10}{2x^2+4x}$

$\frac{5}{x(x+2)}$

e $\frac{x^2-4}{x+2}$

$x-2$

f $\frac{9-x^2}{3-x}$

$3+x$

Key Ideas

1. To simplify a fraction, the numerator and denominator must share a common …… that is not 1.
2. To simplify algebraic fractions, first …… the numerator and denominator, then cancel any …… ……

Identify whether numerator and denominator are negatives of each other

The numerator and denominator are negatives of each other when one can be obtained from the other by factorising out -1. In other words, all terms have the opposite sign.

✗	✓	✓	✓	✗
$\frac{x-3}{x+3}$	$\frac{-x+3}{x+3}$	$\frac{3-x}{x+3}$	$\frac{2x-7}{7-2x}$	$\frac{2x-7}{7+2x}$

Decide whether these numerators and denominators are negative of each other.

a $\frac{5-x}{x+5}$	**b** $\frac{5-x}{x-5}$	**c** $\frac{3x-1}{1-3x}$
d $\frac{3x-1}{1+3x}$	**e** $\frac{-1-3x}{3x+1}$	**f** $\frac{-5x+7}{5x-7}$

Example Simplify algebraic fractions by simple factorisation

$$\frac{3-x}{x-3}$$
$$= \frac{-(x-3)}{x-3}$$
$$= -1$$

$$\frac{9-x^2}{x-3}$$
$$= \frac{(3-x)(3+x)}{x-3}$$
$$= \frac{-(x-3)(3+x)}{x-3}$$

Guided Practice

Simplify

a $\frac{5-x}{x-5}$ -1	**b** $\frac{2x-7}{7-2x}$ -1	**c** $\frac{6-9x}{3x-2}$ -3
d $\frac{5-10x}{6x-3}$ $-\frac{5}{3}$	**e** $\frac{16-x^2}{x-4}$ $-(x+4)$	**f** $\frac{x^2-9}{3-x}$ $-(x+3)$

1 Simplify.

a $\frac{2x+6}{2}$ $x+3$

b $\frac{3x+6}{3}$ $x+2$

c $\frac{4x+6}{2}$ $2x+3$

d $\frac{6x+9}{3}$ $2x+3$

e $\frac{6x+9}{9}$ $\frac{2x+3}{3}$

f $\frac{6x+9}{6}$ $\frac{2x+3}{2}$

2 Simplify.

a $\frac{3x+6}{3}$ $x+2$

b $\frac{3x-6}{3}$ $x-2$

c $\frac{-3x+6}{3}$ $-x+2$

d $\frac{-3x-6}{3}$ $-x-2$

e $\frac{3x+6}{-3}$ $-x-2$

f $\frac{-3x-6}{-3}$ $x+2$

3 Simplify.

a $\frac{4(x+3)}{x+3}$ 4

b $\frac{x+3}{4(x+3)}$ $\frac{1}{4}$

c $\frac{4(x+3)}{8(x+3)}$ $\frac{1}{2}$

d $\frac{4(x+3)^2}{x+3}$ $4(x+3)$

e $\frac{4(x+3)}{(x+3)^2}$ $\frac{4}{x+3}$

f $\frac{(x+3)^2}{4(x+3)}$ $\frac{x+3}{4}$

4 Simplify these expressions by cancelling common factors.

a $\frac{2x}{4}$ $\frac{x}{2}$

b $\frac{6a}{5a}$ $\frac{6}{5}$

c $\frac{4y}{20y}$ $\frac{1}{5}$

d $\frac{3(x+1)}{9(x+1)}$ $\frac{1}{3}$

e $\frac{2(x-2)}{8(x-2)}$ $\frac{1}{4}$

f $\frac{8(x+4)}{12(x+4)}$ $\frac{2}{3}$

g $\frac{(x+5)(x-5)}{(x+5)}$ $x-5$

h $\frac{6(x-1)(x+3)}{9(x+3)}$ $\frac{2(x-1)}{3}$

i $\frac{8(x-2)}{4(x-2)(x+4)}$ $\frac{2}{x+4}$

5 Complete these factorisations.

a $\frac{2x-4}{8} = \frac{2(\quad)}{8}$ $\frac{2(x-2)}{8}$

b $\frac{12-18x}{2x-3x^2} = \frac{6(\quad)}{x(\quad)}$ $\frac{6(2-3x)}{x(2-3x)}$

6 Expand the following.

a $\frac{1}{2}(6y+10)$ — $3y+5$

b $\frac{1}{3}(9x-12)$ — $3x-4$

c $\frac{1}{5}(5s-20t+5)$ — $s-4t+1$

d $-\frac{1}{10}(20-90k)$ — $-2+9k$

7 Simplify, if possible.

a $\frac{3x+6}{3}$ — $x+2$

b $\frac{3x+6}{x+2}$ — 3

c $\frac{3x+6}{x+6}$ — Cannot simplify

d $\frac{3x+6}{6}$ — $\frac{x+2}{2}$

e $\frac{3x+6}{3x}$ — $\frac{x+2}{x}$

f $\frac{3x+6}{9x+18}$ — $\frac{1}{3}$

g $\frac{3x+6}{3x+6}$ — 1

h $\frac{3x+6}{3x+2}$ — Cannot simplify

8 Factorise, then simplify

a $\frac{x^2+3x}{x}$ — $x+3$

b $\frac{x^2-5x}{x}$ — $x-5$

c $\frac{3x^2+6x}{3x}$ — $x+2$

d $\frac{4x^2+8x}{2x}$ — $2(x+2)$

e $\frac{5x^2-10x}{5x}$ — $x-2$

f $\frac{ax+bx}{x}$ — $a+b$

g $\frac{ax^2+bx}{x}$ — $ax+b$

h $\frac{ax^2+bx}{ax}$ — $x+\frac{b}{a}$

9 Simplify, if possible.

a $\frac{5x+10}{5}$ — $x+2$

b $\frac{5x+10}{x+2}$ — 5

c $\frac{5x+10}{x+10}$ — Cannot simplify

d $\frac{x^2+4x}{4x}$ — $\frac{x+4}{4}$

e $\frac{x^2+4x}{x}$ — $x+4$

f $\frac{x^2+4}{x}$ — Cannot simplify

g $\frac{x^2+4}{x+2}$ — Cannot simplify

h $\frac{x^2+4x+4}{x+2}$ — $x+2$

i $\frac{4(2x-1)}{2(1-2x)}$ — -2

10 Kylie simplifies $\frac{4x+6}{2}$ and gets $2x + 6$. Explain her error and give the correct answer.

She divided only the first term by 2. Every term in the numerator must be divided: $\frac{4x+6}{2} = \frac{4x}{2} + \frac{6}{2} = 2x + 3$.

Equivalently, factorise first: $\frac{2(2x+3)}{2} = 2x + 3$.

11 Simplify by factorising before cancelling.

a	$\frac{5x-5}{5}$	$x - 1$	**b**	$\frac{4x-12}{10}$	$\frac{2(x-3)}{5}$
c	$\frac{2x-4}{3x-6}$	$\frac{2}{3}$	**d**	$\frac{15x+9y}{3}$	$5x + 3y$
e	$\frac{6}{12r-18}$	$\frac{1}{2r-3}$	**f**	$\frac{7-y}{y-7}$	-1
g	$\frac{x^2-3x}{x}$	$x - 3$	**h**	$\frac{4x^2+10x}{5x}$	$\frac{2(2x+5)}{5}$
i	$\frac{3x+3y}{2x+2y}$	$\frac{3}{2}$	**j**	$\frac{x^2+x}{x}$	$x + 1$
k	$\frac{x^2-2x}{x}$	$x - 2$	**l**	$\frac{2x-4xy}{2x}$	$1 - 2y$
m	$\frac{2x+4}{5x+10}$	$\frac{2}{5}$	**n**	$\frac{7x-7}{2x-2}$	$\frac{7}{2}$
o	$\frac{3ac+5a}{a+2ab}$	$\frac{3c+5}{1+2b}$	**p**	$\frac{7a-21}{2a-6}$	$\frac{7}{2}$
q	$\frac{12p}{80+2pq}$	$\frac{6p}{40+pq}$	**r**	$\frac{100-10x}{20-2x}$	5

12 This is a common mistake when cancelling. Explain why it is incorrect.

$$\frac{2x + 4^2}{2^1} = 2x + 2$$ ✗

You must divide every term in the numerator by 2.

13 Rewrite each division as multiplication by a fraction, then expand. Simplify the answer.

a $\frac{2y+8}{2} = \frac{1}{2}(2y+8)$

$y+4$

b $\frac{12-16y}{4}$

$3-4y$

c $\frac{18k-6}{6}$

$3k-1$

d $\frac{30t+15}{5}$

$6t+3$

e $\frac{63-14k}{-7}$

$-9+2k$

f $\frac{4a+8}{4}+3$

$a+5$

g $2y+\frac{4y-16}{4}$

$3y-4$

h $\frac{60y-24}{12}-2(y+3)$

$3y-8$

14 Simplify, by factoring out -1.

a $\frac{x-4}{4-x}$

-1

b $\frac{4-x}{x-4}$

-1

c $\frac{2x-5}{5-2x}$

-1

d $\frac{5-2x}{2x-5}$

-1

e $\frac{3-x}{x-3}$

-1

f $\frac{a-b}{b-a}$

-1

g $\frac{2(x-3)}{6-2x}$

-1

h $\frac{3(x-2)}{4(2-x)}$

$-\frac{3}{4}$

i $\frac{6(1-x)}{4(x-1)}$

$-\frac{3}{2}$

15 A student writes $\frac{x+3}{x+5} = \frac{3}{5}$ by "cancelling the x".

Explain why this is wrong and show a numerical counterexample.

Only common factors of the entire numerator and denominator can be cancelled, not individual terms.

Counterexample: let $x = 2$. Left side: $\frac{5}{7}$. Right side: $\frac{3}{5}$. These are not equal.

16 The expression $\frac{6x+k}{3}$ simplifies to $2x + 4$. Find k.

$\frac{6x+k}{3} = 2x + \frac{k}{3}$. So $\frac{k}{3} = 4$, giving $k = 12$

17 For what value of n does $\frac{nx-6}{x-2} = 3$?

The numerator must equal $3(x - 2) = 3x - 6$. So $n = 3$.

18 Is the following statement always, sometimes, or never true? Explain.

$$\frac{a+b}{a} = 1 + \frac{b}{a}$$

Always true for when $a \neq 0$.

Splitting the fraction: $LHS = \frac{a+b}{a} = \frac{a}{a} + \frac{b}{a} = 1 + \frac{b}{a} = RHS$

19 A path has area $(10x - 15)$ m² and length $(2x - 3)$ m.
Show that the width of the path is constant and state its value.

Width $= \frac{10x-15}{2x-3} = \frac{5(2x-3)}{2x-3} = 5$ m. The width is 5 m regardless of x.

20 Simplify the following by first factorising using the difference of two squares.

a $\frac{x^2-100}{x+10}$

$x - 10$

b $\frac{x^2-49}{x+7}$

$x - 7$

c $\frac{x^2-25}{x+5}$

$x - 5$

d $\frac{2(x-20)}{x^2-400}$

$\frac{2}{x+20}$

e $\frac{5(x-6)}{x^2-36}$

$\frac{5}{x+6}$

f $\frac{3x+27}{x^2-81}$

$\frac{3}{x-9}$

21 Factorise the following and simplify.

a $\frac{7-3x}{3x-7}$ -1	**b** $\frac{4x-1}{1-4x}$ -1	**c** $\frac{8x+16}{-2-x}$ -8
d $\frac{x+3}{-9-3x}$ $-\frac{1}{3}$	**e** $\frac{5-3x}{18x-30}$ $-\frac{1}{6}$	**f** $\frac{x^2-9}{3-x}$ $-(x+3)$
g $\frac{-x^2-4x}{-4x}$ $\frac{x+4}{4}$	**h** $\frac{5x-2x^2}{2x^2-5x}$ -1	**i** $\frac{5y-3xy}{18xy-30y}$ $-\frac{1}{6}$
j $\frac{x^2-100}{10-x}$ $-(10+x)$	**k** $\frac{-5a-20}{4+a}$ -5	**l** $\frac{x^2-49}{35-5x}$ $-\frac{x+7}{5}$

Simplifying Algebraic Fractions by Further Factorisation

Review

Factorise

a $x^2 + 5x + 6$	**b** $x^2 - 3x - 10$
c $x^2 - 49$	**d** $x^2 - x + 6$

Simplify fractions by cancelling binomial factors

Simplify these fractions by cancelling common factors.

a $\frac{(x+1)(x+4)}{(x+1)}$	**b** $\frac{(x-3)(x+2)}{(x+2)(x+5)}$	**c** $\frac{2(x+3)(x-1)}{8(x+3)(x+2)}$
d $\frac{(x+3)(x+4)}{(x+4)}$	**e** $\frac{2(x-1)(x+5)}{4(x+5)}$	**f** $\frac{(x-3)(x+2)}{3(x+2)(x+7)}$
g $\frac{x-2}{(2-x)(x+3)}$	**h** $\frac{(x-5)(x+7)}{(x+5)(5-x)}$	**i** $\frac{3(x-4)^2}{(x-4)(x+2)}$

Factorising Algebraic Fractions by Further Factorisation

1. Factorise the numerator and denominator.
2. Cancel any common factors.

Example Factorise Algebraic Fractions by Further Factorisation

$$\frac{x^2-x}{x^2+2x-3}$$

1. $\frac{x(x-1)}{(x+3)(x-1)}$
2. $\frac{x}{x+3}$

$$\frac{x^2-1}{2x^2+4x+2}$$

1. $\frac{(x-1)(x+1)}{2(x^2+2x+1)}$

   $\frac{(x-1)(x+1)}{2(x+1)^2}$
2. $\frac{x-1}{2(x+1)}$

Guided Practice

Simplify

a $\frac{x^2+5x+6}{x+2}$

$x+3$

b $\frac{x^2+6x+8}{x+4}$

$x+2$

c $\frac{a^2-9}{a^2+a-12}$

$\frac{a+3}{a+4}$

d $\frac{x^2+11x+28}{x^2+7x+12}$

$\frac{x+7}{x+3}$

1 Simplify

a $\frac{x^2+5x+4}{x+1}$ $x+4$

b $\frac{x^2+5x+4}{x+4}$ $x+1$

c $\frac{x^2+5x+6}{x+2}$ $x+3$

d $\frac{x^2+5x+6}{x+3}$ $x+2$

e $\frac{x^2+7x+12}{x+3}$ $x+4$

f $\frac{x^2+7x+12}{x+4}$ $x+3$

2 Simplify

a $\frac{x^2+5x+6}{x+2}$ $x+3$

b $\frac{x^2+x-6}{x+3}$ $x-2$

c $\frac{x^2-5x+6}{x-2}$ $x-3$

d $\frac{x^2-x-6}{x-3}$ $x+2$

e $\frac{x^2-x-6}{x+2}$ $x-3$

f $\frac{x^2+x-6}{x-2}$ $x+3$

3 Simplify

a $\frac{x^2+5x+6}{x+2}$ $x+3$

b $\frac{x+3}{x^2+5x+6}$ $\frac{1}{x+2}$

c $\frac{x^2+5x+6}{x^2+7x+12}$ $\frac{x+2}{x+4}$

d $\frac{x^2+7x+12}{x^2+5x+6}$ $\frac{x+4}{x+2}$

4 Simplify

a $\frac{x^2-4}{x+2}$

$x-2$

b $\frac{x^2-4}{x-2}$

$x+2$

c $\frac{x^2-9}{x+3}$

$x-3$

d $\frac{x^2-9}{x-3}$

$x+3$

e $\frac{x^2-25}{x+5}$

$x-5$

f $\frac{x^2-25}{x-5}$

$x+5$

g $\frac{4x^2-9}{2x+3}$

$2x-3$

h $\frac{4x^2-9}{2x-3}$

$2x+3$

i $\frac{9x^2-16}{3x+4}$

$3x-4$

5 Simplify

a $\frac{x^2+6x+9}{x+3}$

$x+3$

b $\frac{x^2-6x+9}{x-3}$

$x-3$

c $\frac{x^2+4x+4}{x+2}$

$x+2$

d $\frac{x+3}{x^2+6x+9}$

$\frac{1}{x+3}$

e $\frac{x^2+6x+9}{x^2-9}$

$\frac{x+3}{x-3}$

f $\frac{x^2-9}{x^2+6x+9}$

$\frac{x-3}{x+3}$

g $\frac{x^2-4x+4}{x^2-4}$

$\frac{x-2}{x+2}$

h $\frac{x^2+10x+25}{x^2-25}$

$\frac{x+5}{x-5}$

6 Simplify

a $\frac{9-x^2}{x+3}$

$3-x$

b $\frac{9-x^2}{x-3}$

$-(x+3)$

c $\frac{x^2-9}{3-x}$

$-(x+3)$

d $\frac{16-x^2}{x-4}$

$-(x+4)$

e $\frac{16-x^2}{x+4}$

$-(x-4)$

f $\frac{x^2-16}{4-x}$

$-(x+4)$

g $\frac{25-4x^2}{2x-5}$

$-(2x+5)$

h $\frac{25-4x^2}{2x+5}$

$-(2x-5)$

7 Simplify

a $\frac{x^2+5x+6}{x+2}$

$x+3$

b $\frac{x+2}{x^2+5x+6}$

$\frac{1}{x+3}$

c $\frac{x^2-9}{x-3}$

$x+3$

d $\frac{x-3}{x^2-9}$

$\frac{1}{x+3}$

e $\frac{x^2-9}{x+3}$

$x-3$

f $\frac{x+3}{x^2-9}$

$\frac{1}{x-3}$

8 A student simplifies $\frac{x^2+4}{x+2}$ and writes the answer as $x+2$. Explain the error.

$x^2+4 \neq (x+2)^2$.

The expression x^2+4 cannot be factorised over the real numbers, so $(x+2)$ is not a factor of the numerator and cancellation is not possible.

9 Simplify these expressions.

a $\dfrac{(a+1)^2}{a+1}$

$a+1$

b $\dfrac{5(a-3)^2}{a-3}$

$5(a-3)$

c $\dfrac{7(x+7)^2}{14(x+7)}$

$\dfrac{x+7}{2}$

d $\dfrac{3(x-1)(x+2)^2}{18(x-1)(x+2)}$

$\dfrac{x+2}{6}$

e $\dfrac{x^2+6x+9}{2x+6}$

$\dfrac{x+3}{2}$

f $\dfrac{11x-22}{x^2-4x+4}$

$\dfrac{11}{x-2}$

g $\dfrac{x^2-6x+9}{x-3}$

$x-3$

h $\dfrac{4x+20}{x^2+10x+25}$

$\dfrac{4}{x+5}$

i $\dfrac{x^2-14x+49}{5x-35}$

$\dfrac{x-7}{5}$

10 Simplify

a $\dfrac{9-x^2}{x+3}$

$3-x$

b $\dfrac{9-x^2}{x-3}$

$-(x+3)$

c $\dfrac{x^2-9}{3-x}$

$-(x+3)$

d $\dfrac{16-x^2}{x-4}$

$-(x+4)$

e $\dfrac{16-x^2}{x+4}$

$-(x-4)$

f $\dfrac{x^2-16}{4-x}$

$-(x+4)$

g $\dfrac{25-4x^2}{2x-5}$

$-(2x+5)$

h $\dfrac{25-4x^2}{2x+5}$

$-(2x-5)$

11 Explain why $\frac{x^2-9}{x-3}$ is not defined when $x = 3$.

The original expression has $(x-3)$ in the denominator, so it is undefined at $x = 3$ (division by zero).

The simplified form $x+3$ does not carry this restriction, so the two expressions are equivalent only for $x \neq 3$.

12 A square has area $(x^2 - 12x + 36)$ cm². Write an expression for its side length.

Area of square = side^2, so side = $\sqrt{x^2 - 12x + 36} = \sqrt{(x-6)^2} = x - 6$ cm (for $x > 6$).

13 Simplify by first factorising.

a $\dfrac{x^2-x-6}{x-3}$

$x + 2$

b $\dfrac{x^2+8x+16}{x+4}$

$x + 4$

c $\dfrac{x^2-7x+12}{x-4}$

$x - 3$

d $\dfrac{x-2}{x^2+x-6}$

$\dfrac{1}{x+3}$

e $\dfrac{x+7}{x^2+5x-14}$

$\dfrac{1}{x-2}$

f $\dfrac{x-9}{x^2-19x+90}$

$\dfrac{1}{x-10}$

g $\dfrac{c^2+11c+30}{6c+30}$

$\dfrac{c+6}{6}$

h $\dfrac{he-hd+pe-pd}{e^2-d^2}$

$\dfrac{h+p}{e+d}$

i $\dfrac{k^2+5k-24}{2k^2-18}$

$\dfrac{k+8}{2(k+3)}$

j $\dfrac{r^2-2r-24}{r^2+12r+32}$

$\dfrac{r-6}{r+8}$

k $\dfrac{6m-30-pm+5p}{m^2+3m-40}$

$\dfrac{6-p}{m+8}$

l $\dfrac{n^3-5n^2+4n}{n^2-2n+1}$

$\dfrac{n(n-4)}{n-1}$

14 Simplify these expressions.

a $\frac{2x}{x}$

2

b $\frac{2x+7}{x}$

can't simplify

c $\frac{2x+7}{7}$

can't simplify

d $\frac{2(x+3)}{3(x+3)}$

$\frac{2}{3}$

e $\frac{2(x+3)}{(x+3)}$

2

f $\frac{(x+3)}{2(x+3)}$

$\frac{1}{2}$

g $\frac{x+3}{2x+6}$

$\frac{1}{2}$

h $\frac{(x+3)(x+4)}{2x+6}$

$\frac{x+4}{2}$

i $\frac{(x+3)(x+4)}{2x+8}$

$\frac{x+3}{2}$

j $\frac{x^2+7x+12}{2x+8}$

$\frac{x+3}{2}$

k $\frac{x^2+7x+12}{(x+4)(x-9)}$

$\frac{x+3}{x-9}$

l $\frac{x^2+7x+12}{x^2-5x-36}$

$\frac{x+3}{x-9}$

m $\frac{2x^2+14x+24}{x^2-5x-36}$

$\frac{2(x+3)}{x-9}$

n $\frac{2x^2+14x+24}{3x^2-15x-108}$

$\frac{2(x+3)}{3(x-9)}$

o $\frac{2x^2+14x+24}{3x^2+4x-15}$

$\frac{2(x+4)}{3x-5}$

p $\frac{14x-24-2x^2}{3x^2+4x-15}$

$-\frac{2(x-3)(x-4)}{(3x-5)(x+3)}$

q $\frac{14x-24-2x^2}{3x^2-4x-15}$

$-\frac{2(x-4)}{3x+5}$

15 Find values of a and b such that $\frac{x^2+ax+b}{x+3} = x - 4$ for all valid x.

Multiply both sides by $(x+3)$: $x^2 + ax + b = (x+3)(x-4) = x^2 - x - 12$.

So $a = -1$, $b = -12$.

16 The expression $\frac{x^2+5x+6}{x^2+ax+b}$ simplifies to $\frac{x+2}{x+5}$. Find a and b.

Numerator $= (x+2)(x+3)$. The factor $(x+3)$ cancels, so the denominator must have $(x+3)$ as a factor and the remaining factor is $(x+5)$.
Denominator $= (x+3)(x+5) = x^2 + 8x + 15$. So $a = 8$, $b = 15$.

17 For which integer values of k does $\frac{x^2+kx+12}{x+3}$ simplify to a linear expression?

$(x+3)$ must be a factor of $x^2 + kx + 12$. Write $x^2 + kx + 12 = (x+3)(x+c)$ where $3c = 12$, so $c = 4$ and $k = 3 + 4 = 7$.

18 Simplify by factorising using the difference of two squares first.

a $\frac{x^2-7}{x+\sqrt{7}}$

$x - \sqrt{7}$

b $\frac{x^2-10}{x-\sqrt{10}}$

$x + \sqrt{10}$

c $\frac{\sqrt{5}x+3}{5x^2-9}$

$\frac{1}{\sqrt{5}-3}$

d $\frac{\sqrt{3}x-4}{3x^2-16}$

$\frac{1}{\sqrt{3}x+4}$

e $\frac{(x+1)^2-2}{x+1+\sqrt{2}}$

$x + 1 - \sqrt{2}$

f $\frac{(x-3)^2-5}{x-3-\sqrt{5}}$

$x - 3 + \sqrt{5}$

19 Simplify by first factorising.

a $\dfrac{6x^2-x-35}{3x+7}$

$2x-5$

b $\dfrac{8x^2+10x-3}{2x+3}$

$4x-1$

c $\dfrac{10x-2}{15x^2+7x-2}$

$\dfrac{2}{3x+2}$

d $\dfrac{20x-12}{10x^2-21x+9}$

$\dfrac{4}{2x-3}$

e $\dfrac{2x^2+11x+12}{6x^2+11x+3}$

$\dfrac{x+4}{3x+1}$

f $\dfrac{12x^2-x-1}{8x^2+14x+3}$

$\dfrac{3x-1}{2x+3}$

g $\dfrac{9x^2-4}{15x^2+4x-4}$

$\dfrac{3x-2}{5x-2}$

h $\dfrac{14x^2+19x-3}{49x^2-1}$

$\dfrac{2x+3}{7x+1}$

i $\dfrac{8x^2-2x-15}{16x^2-25}$

$\dfrac{2x-3}{4x-5}$

j $\dfrac{25-4a^2}{6a^2-11a-10}$

$-\dfrac{2a+5}{3a+2}$

k $\dfrac{40+11b-2b^2}{8b^2+34b+35}$

$\dfrac{8-b}{4b+7}$

l $\dfrac{x^2y^2-4}{xy^2-3mxy-2y+6m}$

$\dfrac{xy-2}{y-3m}$

Multiplying and Dividing Algebraic Fractions by Factorising

Simplify fractions by cancelling factors

Simplify these fractions by cancelling common factors.

a

$$\frac{(x+2)(x-3)}{(x+1)(x+2)} \times \frac{(x+1)(x+5)}{(x-3)(x+4)}$$

b

$$\frac{3(x-1)}{(x+2)(x-1)} \times \frac{(x+2)(x+3)}{6}$$

c

$$\frac{(x+4)(x-2)}{(x+3)(x+4)} \div \frac{(x-2)}{(x+3)(x+1)}$$

d

$$\frac{(x+1)(x-2)}{(x+3)(x+1)} \times \frac{(x+3)(x+5)}{(x-2)(x+7)}$$

e

$$\frac{2(x+4)}{(x-1)(x+4)} \div \frac{6}{(x-1)}$$

f

$$\frac{(x+1)}{(1-x)} \div \frac{(x+5)}{(x-1)}$$

Multiplying and Dividing Fractions by Factorising

0. **Change division to multiplication** by the reciprocal.
1. **Factorise** all numerators and denominators.
2. **Cancel** any common factors vertically or diagonally.
3. **Multiply** the numerators and multiply the denominators.

Example Multiply and divide algebraic fractions by factorising

$$\frac{2a}{a^2-9}\times\frac{a-3}{5a}$$

1. $\frac{2a}{(a-3)(a+3)}\times\frac{a-3}{5a}$
2. $\frac{2a}{a+3}\times\frac{1}{5a}$
3. $\frac{2}{a+3}\times\frac{1}{5}$

$$\frac{2}{5(a+3)}$$

Guided Practice

Simplify

a $\frac{3x+6}{5}\times\frac{10}{x+2}$

$=\frac{3(x+2)}{5}\times\frac{10}{x+2}$

$=$

6

b $\frac{3x+3}{2x}\times\frac{x^2}{x^2-1}$

$=\frac{3(\quad)}{2x}\times\frac{x^2}{(\quad)(\quad)}$

$=$

$\frac{3x}{2(x-1)}$

c $\frac{a^2-4}{3}\times\frac{5b}{a+2}$

$\frac{5b(a-2)}{3}$

d $\frac{6x-12}{5}\div\frac{x-2}{3}$

$\frac{18}{5}$

e $\frac{3-6y}{8}\div\frac{1-2y}{2}$

$\frac{3}{4}$

f $\frac{12x}{x+1}\div\frac{6x}{x^2+2x+1}$

$2(x+1)$

1 Simplify

a $\frac{x+1}{x+2} \times \frac{x+2}{x+3}$

$\frac{x+1}{x+3}$

b $\frac{x+1}{x+2} \times \frac{x+2}{x+4}$

$\frac{x+1}{x+4}$

c $\frac{x+1}{x+2} \times \frac{x+3}{x+1}$

$\frac{x+3}{x+2}$

d $\frac{x+2}{x+3} \times \frac{x+3}{x+2}$

1

e $\frac{x-1}{x+2} \times \frac{x+2}{x-1}$

1

f $\frac{x+1}{x+2} \times \frac{x+4}{x+3}$

$\frac{(x+1)(x+4)}{(x+2)(x+3)}$

2 Simplify

a $\frac{x+3}{2} \times \frac{4}{x+3}$

2

b $\frac{x+3}{2} \div \frac{x+3}{4}$

2

c $\frac{x+3}{2} \times \frac{2}{x+3}$

1

d $\frac{x+3}{4} \div \frac{x+3}{2}$

$\frac{1}{2}$

e $\frac{x+3}{4} \times \frac{4}{x+5}$

$\frac{x+3}{x+5}$

f $\frac{x+3}{4} \div \frac{x+5}{4}$

$\frac{x+3}{x+5}$

3 Simplify

a $\frac{x^2-4}{x+2} \times \frac{1}{x-2}$

1

b $\frac{1}{x^2-4} \times (x+2)$

$\frac{1}{x-2}$

c $(x^2-4) \times \frac{1}{x+2}$

$x-2$

d $\frac{x^2-9}{x+3} \div (x-3)$

1

e $(x^2-9) \div \frac{x+3}{x-3}$

$(x-3)^2$

f $\frac{x+3}{x^2-9} \times (x-3)$

1

4 Simplify

a $\frac{x+1}{x+2} \times \frac{x+2}{x+3} \times \frac{x+3}{x+4}$

$\frac{x+1}{x+4}$

b $\frac{x+1}{x+2} \times \frac{x+2}{x+3} \times \frac{x+4}{x+1}$

$\frac{x+4}{x+3}$

c $\frac{x^2-1}{x+2} \times \frac{x+2}{x^2-4} \times \frac{x-2}{x+1}$

$\frac{x-1}{x+2}$

d $\frac{a^2-b^2}{a+c} \times \frac{a+c}{a-b} \times \frac{1}{a+b}$

1

5 Simplify.

a $\frac{2x(x-4)}{4(x+1)} \times \frac{x+1}{x}$

$\frac{x-4}{2}$

b $\frac{(x+2)(x-3)}{x-5} \times \frac{x-5}{x+2}$

$x-3$

c $\frac{x-3}{x+2} \times \frac{3(x+4)(x+2)}{x+4}$

$3(x-3)$

6 Simplify.

a $\frac{x(x+1)}{(x+3)} \div \frac{x+1}{x+3}$

x

b $\frac{x+3}{x+2} \div \frac{x+3}{2(x-2)}$

$\frac{2(x-2)}{x+2}$

c $\frac{5(2x-3)}{x+7} \div \frac{(x+2)(2x-3)}{x+7}$

$\frac{5}{x+2}$

7 Simplify

a $\frac{x^2-1}{x+1} \div \frac{x-1}{3}$

3

b $\frac{x^2-4}{x+2} \div (x-2)$

1

c $\frac{x^2+5x+6}{x+3} \div \frac{x+2}{4}$

4

d $\frac{x^2+5x+6}{x^2-x-6} \div \frac{x+2}{x-3}$

$\frac{x+3}{x+2}$

e $\frac{x^2-16}{x^2-5x+4} \div \frac{x+4}{x-1}$

1

f $\frac{(x-3)(x+3)}{2} \times \frac{6}{x+3}$

$3(x-3)$

8

9 A runner's speed is $\frac{x+3}{x+1}$ m/s and she runs for $\frac{x^2-1}{x+3}$ seconds.
Write an expression for the distance covered.

$\text{Distance} = \frac{x+3}{x+1} \times \frac{(x-1)(x+1)}{x+3} = x-1$ m.

10 A tank fills at a rate of $\frac{x^2+3x}{x+3}$ litres per minute. The tank holds $\frac{x^2+3x+2}{x+1}$ litres.
Write an expression for how long it takes to fill.

$\text{Time} = \text{volume} \div \text{rate}$

$= \frac{(x+1)(x+2)}{x+1} \times \frac{x+3}{x(x+3)}$

$= \frac{x+2}{x}$ minutes.

11 Simplify.

a $\frac{x^2-4}{x^2-x-6} \times \frac{5x-15}{x^2+4x-12}$

$\frac{5}{x+6}$

b $\frac{x^2+3x+2}{x^2+4x+3} \times \frac{x^2-9}{3x+6}$

$\frac{x-3}{3}$

c $\frac{x^2+2x-3}{x^2-25} \times \frac{2x-10}{x+3}$

$\frac{2(x-1)}{x+5}$

d $\frac{x^2-4x+3}{x^2+4x-21} \times \frac{5x-15}{x^2+4x-12}$

$\frac{5(x-3)(x-1)}{(x+7)(x+6)(x-2)}$

e $\frac{x^2-x-6}{x^2+x-12} \times \frac{x^2+5x+4}{x^2-1}$

$\frac{x+2}{x-1}$

f $\frac{x^2-4x-12}{x^2-4} \times \frac{x^2-6x+8}{x^2-36}$

$\frac{x-4}{x+6}$

g $\frac{x^2+2x-3}{x^2-25} \div \frac{3x-3}{2x+10}$

$\frac{2(x+3)}{3(x-5)}$

h $\frac{x^2+3x+2}{x^2+4x+3} \div \frac{4x+8}{x^2-9}$

$\frac{x-3}{4}$

i $\frac{x^2+5x-14}{x^2+2x-3} \div \frac{x^2+9x+14}{x^2+x-2}$

$\frac{x-2}{x+3}$

12 Find an expression for the area of these rectangles.

a



$\frac{x+1}{2}$ m

$x + 1$ m^2

b



$\frac{4}{x-3}$ cm

2 cm^2

c



$\frac{2x-4}{5}$ mm

4 mm^2

13 Without fully simplifying, explain why $\frac{x^2-9}{x+3} \times \frac{x+3}{x^2-9} = 1$ for valid values of x.

The two fractions are reciprocals of each other. Any non-zero quantity multiplied by its reciprocal equals 1.

14 The product $\frac{x^2+bx+c}{x^2-9} \times \frac{x-3}{x+2}$ simplifies to $\frac{x+5}{x+2}$. Find b and c.

After simplification the $(x - 3)$ in the numerator must cancel with a factor in $x^2 - 9 = (x - 3)(x + 3)$, leaving $(x + 3)$ in the denominator.

Then the numerator $x^2 + bx + c$ divided by $(x + 3)$ must give $(x + 5)$,

so $x^2 + bx + c = (x + 3)(x + 5) = x^2 + 8x + 15$.

Thus $b = 8$, $c = 15$.

15 Find the missing algebraic fraction.

a $\frac{x+3}{5} \times \square = 2$

$\frac{10}{x+3}$

b $\frac{1-x}{x} \times \square = 3$

$\frac{3x}{1-x}$

c $\square \div \frac{x}{2} = \frac{3(x+2)}{x}$

$\frac{3(x+2)}{2}$

d $\square \div \frac{2x-2}{3} = \frac{5x}{x-1}$

$\frac{10x}{3}$

e $\frac{1}{x} \div \square \times \frac{x-1}{2} = 1$

$\frac{x-1}{2x}$

f $\frac{2-x}{7} \times \square \div \frac{5x}{x-1} = x$

$\frac{35x^2}{(x-1)(2-x)}$

16 Simplify

a $\dfrac{a^2+2ab+b^2}{a(a+b)} \div \dfrac{a^2-b^2}{a^2-2ab+b^2}$

$\frac{a-b}{a}$

b $\dfrac{a^2-2ab+b^2}{a^2-b^2} \div \dfrac{a^2-b^2}{a^2+2ab+b^2}$

1

c $\dfrac{a^2-b^2}{a^2-2ab+b^2} \div \dfrac{a^2-b^2}{a^2+2ab+b^2}$

$\frac{(a+b)^2}{(a-b)^2}$

d $\dfrac{a^2+2ab+b^2}{a(a+b)} \div \dfrac{a(a-b)}{a^2-2ab+b^2}$

$\frac{(a+b)(a-b)}{a^2}$

17 Simplify these expressions.

e $\dfrac{x^2-x-20}{x^2-25} \times \dfrac{x^2-x-2}{x^2+2x-8} \div \dfrac{x+1}{x^2+5x}$

x

f $\dfrac{x^2-8x+15}{5x^2+10x} \div \dfrac{x^2-9}{10x^2} \times \dfrac{x^2+5x+6}{2x-10}$

x

g $\dfrac{100x^2-25}{2x^2-9x-5} \div \dfrac{5x^2+10x-75}{2x^2-5x-3}$

$\frac{5(2x-1)(2x+1)}{(x-5)(x+5)}$

h $\dfrac{3x^2-12}{30x+15} \div \dfrac{2x^2-3x-2}{4x^2+4x+1}$

$\frac{x+2}{5}$

Adding and Subtracting Algebraic Fractions by Factorising

Identify lowest common multiple of two algebraic terms

Identify the lowest common multiple of these terms.

a	$6x \ \& \ 9x^2$	**b**	$(x-2) \ \& \ 4(x-2)$
c	$(x+3)(x-1) \ \& \ (x-1)(x+7)$	**d**	$x(x+4) \ \& \ (x+4)(x-2)$
e	$(x+1)^2 \ \& \ (x+1)(x-5)$	**f**	$(x+2)(x-3) \ \& \ (x-3)(2x+1)$
g	$x(x-2) \ \& \ (x-2)(x+3)$	**h**	$4x(x+5) \ \& \ 2x^2(x+5)(x-3)$
i	$x+2 \ \& \ x^2+5x+6$	**j**	$x^2-4 \ \& \ x+2$

Adding and Subtracting Fractions by Factorising

1. **Factorise** the denominators.
2. Change denominators to the **Lowest Common Denominator** using equivalent fractions.
3. Add or subtract the numerators.

Note: All fractions have invisible brackets in the numerator and denominator $\frac{a+b}{c+d} = \frac{(a+b)}{(c+d)}$

Example Add and subtract algebraic fractions by factorising

$$\frac{1}{x-4}+\frac{1}{x}$$

2. $\frac{}{x(x-4)}+\frac{}{x(x-4)}$

$$\frac{x}{x(x-4)}+\frac{x-4}{x(x-4)}$$

3. $\frac{2x-4}{x(x-4)}$

$$\frac{2+x}{x^2-x}-\frac{5}{x-1}$$

1. $\frac{2+x}{x(x-1)}-\frac{5}{x-1}$

2. $\frac{}{x(x-1)}-\frac{}{x(x-1)}$

$$\frac{2+x}{x(x-1)}-\frac{5x}{x(x-1)}$$

3. $\frac{2-4x}{x(x-1)}$

Guided Practice

Simplify these expressions.

a $\frac{1}{x}+\frac{1}{x+1}$

$=\frac{}{x(x+1)}+\frac{}{x(x+1)}$

$\frac{2x+1}{x(x+1)}$

b $\frac{4}{x+1}+\frac{3}{x-2}$

$\frac{7x-5}{(x+1)(x-2)}$

c $\frac{2}{x-3}-\frac{3}{x-2}$

$\frac{-x+5}{(x-3)(x-2)}$

d $\frac{3}{(x-1)^2}-\frac{2}{x-1}$

$\frac{5-2x}{(x-1)^2}$

e $\frac{2x}{x-5}-\frac{x}{x+1}$

$\frac{x^2+7x}{(x-5)(x+1)}$

f $\frac{2}{x^2-4}-\frac{3}{x+2}$

$\frac{8-3x}{(x-2)(x+2)}$

1 Identify the LCM

a $(x+1)$ & $(x+2)$

$(x+1)(x+2)$

b $(x+1)$ & $(x+3)$

$(x+1)(x+3)$

c $2(x+1)$ & $(x+2)$

$2(x+1)(x+2)$

d $3(x+1)$ & $2(x+2)$

$6(x+1)(x+2)$

e $(x+1)^2$ & $(x+2)$

$(x+1)^2(x+2)$

f $(x+1)^2$ & $(x+1)(x+2)$

$(x+1)^2(x+2)$

g $(x+2)(x-3)$ & $(x-3)(x+5)$

$(x+2)(x-3)(x+5)$

h $(x+2)(x-3)$ & $(x-3)^2$

$(x+2)(x-3)^2$

i $(x-3)(x+1)$ & $(x-3)(x+4)$

$(x-3)(x+1)(x+4)$

j $2(x+1)(x-4)$ & $3(x-4)(x+2)$

$6(x+1)(x-4)(x+2)$

k x^2-4 & $x+2$

$(x+2)(x-2)$

l x^2-4 & $x-2$

$(x+2)(x-2)$

m x^2-4 & x^2+4x+4

$(x+2)^2(x-2)$

n x^2+5x+6 & x^2+4x+3

$(x+1)(x+2)(x+3)$

o x^2-9 & x^2-x-6

$(x+3)(x-3)(x+2)$

p x^2-1 & x^2+2x+1

$(x+1)^2(x-1)$

q $x^2+7x+12$ & x^2+6x+8

$(x+2)(x+3)(x+4)$

r x^2-4x+4 & x^2-5x+6

$(x-2)^2(x-3)$

2 Simplify.

a $\frac{1}{x+1}+\frac{1}{x+2}$

$\frac{2x+3}{(x+1)(x+2)}$

b $\frac{1}{x+1}+\frac{1}{x+3}$

$\frac{2(x+2)}{(x+1)(x+3)}$

c $\frac{1}{x+1}+\frac{1}{x+2}$

$\frac{2x+3}{(x+1)(x+2)}$

d $\frac{1}{x+1}+\frac{1}{x+4}$

$\frac{2x+5}{(x+1)(x+4)}$

e $\frac{2}{x+1}+\frac{1}{x+2}$

$\frac{3x+5}{(x+1)(x+2)}$

f $\frac{1}{x+1}+\frac{1}{x+4}$

$\frac{2x+5}{(x+1)(x+4)}$

g $\frac{3}{x+1}+\frac{1}{x+2}$

$\frac{4x+7}{(x+1)(x+2)}$

h $\frac{4}{x+1}+\frac{1}{x+2}$

$\frac{5x+9}{(x+1)(x+2)}$

i $\frac{3}{x+1}+\frac{1}{x+2}$

$\frac{4x+7}{(x+1)(x+2)}$

3 Simplify.

a $\frac{1}{x+1}+\frac{1}{x+2}$

$\frac{2x+3}{(x+1)(x+2)}$

b $\frac{1}{x+1}-\frac{1}{x+2}$

$\frac{1}{(x+1)(x+2)}$

c $\frac{1}{x+1}+\frac{1}{x+2}$

$\frac{2x+3}{(x+1)(x+2)}$

d $\frac{2}{x+1}-\frac{1}{x+2}$

$\frac{x+3}{(x+1)(x+2)}$

e $\frac{1}{x+1}-\frac{2}{x+2}$

$-\frac{x}{(x+1)(x+2)}$

f $\frac{2}{x+1}-\frac{1}{x+2}$

$\frac{x+3}{(x+1)(x+2)}$

g $\frac{3}{x+2}-\frac{2}{x+1}$

$\frac{x-1}{(x+1)(x+2)}$

h $\frac{2}{x+2}-\frac{3}{x+1}$

$-\frac{x+4}{(x+1)(x+2)}$

i $\frac{3}{x+2}-\frac{2}{x+1}$

$\frac{x-1}{(x+1)(x+2)}$

4 Simplify the following. Change denominators to be the same first.

a $\frac{5}{x}+\frac{2}{x+4}$

$\frac{7x+20}{x(x+4)}$

b $\frac{4}{x-7}+\frac{3}{x}$

$\frac{7(x-3)}{x(x-7)}$

c $\frac{1}{x-3}+\frac{2}{x+5}$

$\frac{3x-1}{(x-3)(x+5)}$

d $\frac{3}{x+3}-\frac{2}{x-4}$

$\frac{x-18}{(x-4)(x+3)}$

e $\frac{6}{2x-1}-\frac{3}{x-4}$

$\frac{-21}{(x-4)(2x-1)}$

f $\frac{4}{x-5}+\frac{2}{3x-4}$

$\frac{2(7x-13)}{(x-5)(3x-4)}$

g $\frac{5}{2x-1}-\frac{6}{x+7}$

$\frac{41-7x}{(x+7)(2x-1)}$

h $\frac{8}{3x-2}-\frac{3}{1-x}$

$\frac{14-17x}{(1-x)(3x-2)}$

5 The LCM of $x^2 - a^2$ and $x - a$ is $x^2 - a^2$.
What does this tell you about the relationship between these two expressions?

$x - a$ is a factor of $x^2 - a^2$.

6 Simplify

a $\frac{1}{x+2} + \frac{1}{x+3}$

$\frac{2x+5}{(x+2)(x+3)}$

b $\frac{1}{x+2} + \frac{1}{x^2+5x+6}$

$\frac{x+4}{(x+2)(x+3)}$

c $\frac{1}{x+2} + \frac{1}{x+3}$

$\frac{2x+5}{(x+2)(x+3)}$

d $\frac{2}{x^2+5x+6} + \frac{3}{x+2}$

$\frac{3x+11}{(x+2)(x+3)}$

e $\frac{2}{x^2+5x+6} + \frac{3}{x+3}$

$\frac{3x+8}{(x+2)(x+3)}$

f $\frac{2}{x^2+5x+6} + \frac{3}{x+2}$

$\frac{3x+11}{(x+2)(x+3)}$

g $\frac{1}{x^2+5x+6} + \frac{1}{x^2+4x+3}$

$\frac{2x+3}{(x+1)(x+2)(x+3)}$

h $\frac{2}{x^2+3x+2} - \frac{1}{x^2+4x+3}$

$\frac{x+4}{(x+1)(x+2)(x+3)}$

i $\frac{1}{x^2+5x+6} + \frac{1}{x^2+4x+3}$

$\frac{2x+3}{(x+1)(x+2)(x+3)}$

7 Simplify

a $\frac{1}{x-3} + \frac{1}{3-x}$

0

b $\frac{2}{x-3} + \frac{3}{3-x}$

$-\frac{1}{x-3}$

c $\frac{1}{x-3} + \frac{1}{3-x}$

0

d $\frac{x}{x-2} + \frac{4}{2-x}$

$\frac{x-4}{x-2}$

e $\frac{3}{x-1} - \frac{2}{1-x}$

$\frac{5}{x-1}$

f $\frac{x}{x-2} + \frac{4}{2-x}$

$\frac{x-4}{x-2}$

g $\frac{5}{2-x} + \frac{3}{x-2}$

$-\frac{2}{x-2}$

h $\frac{a}{x-b} + \frac{c}{b-x}$

$\frac{a-c}{x-b}$

i $\frac{5}{2-x} + \frac{3}{x-2}$

$-\frac{2}{x-2}$

8 Simplify

a $\frac{1}{x}+\frac{1}{x+1}+\frac{1}{x+2}$

$\frac{3x^2+6x+2}{x(x+1)(x+2)}$

b $\frac{1}{x+1}+\frac{2}{x+2}-\frac{1}{x+3}$

$\frac{2(x^2+5x+5)}{(x+1)(x+2)(x+3)}$

9 Simplify

a $\frac{1}{(x+1)^2}+\frac{1}{x+1}$

$\frac{x+2}{(x+1)^2}$

b $\frac{2}{(x+1)^2}-\frac{1}{x+1}$

$\frac{1-x}{(x+1)^2}$

c $\frac{3}{(x-2)^2}+\frac{2}{x-2}$

$\frac{2x-1}{(x-2)^2}$

d $\frac{x}{(x+3)^2}-\frac{1}{x+3}$

$-\frac{3}{(x+3)^2}$

10 Simplify the following.

a $\frac{4}{(x+1)^2}-\frac{3}{x+1}$

$\frac{1-3x}{(x+1)^2}$

b $\frac{2}{(x+3)^2}-\frac{4}{x+3}$

$\frac{-4x-10}{(x+3)^2}$

c $\frac{3}{x-2}+\frac{4}{(x-2)^2}$

$\frac{3x-2}{(x-2)^2}$

d $\frac{3x}{(x-1)^2}+\frac{2}{x-1}$

$\frac{5x-2}{(x-1)^2}$

e $\frac{3x+2}{3x}+\frac{7}{12}$

$\frac{19x+8}{12x}$

f $\frac{2x-1}{4}+\frac{2-3x}{10x}$

$\frac{10x^2-11x+4}{20x}$

g $\frac{3}{4-x}-\frac{2x}{x-1}$

$\frac{(2x+1)(x-3)}{(4-x)(x-1)}$

h $\frac{x}{x+1}-\frac{5x+1}{(x+1)^2}$

$\frac{x^2-4x-1}{(x+1)^2}$

11 Factorise the denominator before simplifying these fractions.

a $\frac{3}{x+2}+\frac{5}{2x+4}$

$\frac{11}{2(x+2)}$

b $\frac{7}{3x-3}-\frac{2}{x-1}$

$\frac{1}{3(x-1)}$

c $\frac{3}{8x-4}-\frac{5}{1-2x}$

$\frac{23}{4(2x-1)}$

d $\frac{4}{x^2-9}-\frac{3}{x+3}$

$\frac{13-3x}{(x-3)(x+3)}$

e $\frac{5}{2x+4}+\frac{2}{x^2-4}$

$\frac{5x-6}{2(x-2)(x+2)}$

f $\frac{10}{3x-4}-\frac{7}{9x^2-16}$

$\frac{3(10x+11)}{(3x-4)(3x+4)}$

g $\frac{1}{x^2+x}+\frac{1}{x^2-x}$

$\frac{2x}{x(x+1)(x-1)}$

h $\frac{1}{x-y}+\frac{2x-y}{x^2-y^2}$

$\frac{3x}{(x+y)(x-y)}$

i $\frac{3}{x^2+2x-8}-\frac{2}{x^2+x-6}$

$\frac{x+1}{(x-2)(x+3)(x+4)}$

j $\frac{1}{x^2-4x+3}+\frac{1}{x^2-5x+6}-\frac{1}{x^2-3x+2}$

$\frac{x}{(x-3)(x-2)(x-1)}$

12 Simplify

$$\frac{1}{n}+\frac{(n+1)(n-1)}{n}-(n+1)$$

-1

13 Simplify the following.

a $\dfrac{1}{(x+3)(x+4)}+\dfrac{2}{(x+4)(x+5)}$

$\dfrac{3x+11}{(x+3)(x+4)(x+5)}$

b $\dfrac{3}{(x+1)(x+2)}-\dfrac{5}{(x+1)(x+4)}$

$\dfrac{2-2x}{(x+1)(x+2)(x+4)}$

c $\dfrac{4}{(x-1)(x-3)}-\dfrac{6}{(x-1)(2-x)}$

$\dfrac{26-10x}{(x-1)(x-3)(2-x)}$

d $\dfrac{5x}{(x+1)(x-5)}-\dfrac{2}{(x-5)}$

$\dfrac{3x-2}{(x+1)(x-5)}$

e $\dfrac{7}{k(k-1)}-\dfrac{4}{k(k+1)}$

$\dfrac{3k+11}{k(k-1)(k+1)}$

f $\dfrac{8}{(x+4)(x-7)}+\dfrac{3}{x(x+4)}$

$\dfrac{11x-21}{x(x-7)(x+4)}$

14 Explain the mistake.

Express $\dfrac{3}{x^2-2x}+\dfrac{2}{x^2-7x+10}$ as a single fraction in simplest form. **2**

$\dfrac{3(x^2-7x+10)+2(x^2-2x)}{(x^2-2x)(x^2-7x+10)} = \dfrac{(3x^2-21x+30)+(2x^2-4x)}{(x^2-2x)(x^2-7x+10)}$

$= \dfrac{6x^4-12x^3-42x^3-84x^2+60x^2-}{(x^2-2x)(x^2-7x+10)}$

$= \dfrac{6x^4-54x^3-24x^2-120x}{(x^2-2x)(x^2-7x+10)}$

They should have factorised before trying to make a common denominator

Simplifying Compound Fractions

Identify main numerator and main denominator

A compound fraction has a fraction in its numerator, denominator, or both.

Dealing with these fractions requires identifying the ‘main’ numerator and ‘main’ denominator.

a $\dfrac{\frac{1}{x}}{\frac{2}{x+1}}$	**b** $\dfrac{\frac{3}{x}}{1-\frac{5}{x}}$	**c** $\dfrac{1-\frac{2}{x}}{3}$
d $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x}}$	**e** $\dfrac{2+\frac{1}{x+1}}{3-\frac{1}{x+1}}$	**f** $\dfrac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}-\frac{1}{y}}$

Simplifying Compound Fractions

1. Identify the main numerator and main denominator.
2. Multiply every single term in the main numerator and denominator by the LCD.
3. Simplify.

This is the fastest method. An alternate method is:

1. Create a single fraction in the main numerator and denominator.
2. Divide the numerator by the denominator.
3. Simplify.

Example Simplify expressions involving compound fractions

$$\frac{1+\frac{1}{3x}}{1-\frac{1}{x^2}} \quad \begin{matrix}\times 3x^2 \\ \hline \times 3x^2\end{matrix}$$

$$= \frac{3x^2\left(1+\frac{1}{3x}\right)}{3x^2\left(1-\frac{1}{x^2}\right)}$$

$$= \frac{3x^2+x}{3x^2-3}$$

$$= \frac{x(3x+1)}{3(x-1)(x+1)}$$

$$\frac{1+\frac{1}{3x}}{1-\frac{1}{x^2}}$$

$$= \frac{\left(\frac{3x+1}{3x}\right)}{\left(\frac{x^2-1}{x^2}\right)} = \frac{3x+1}{3x} \div \frac{x^2-1}{x^2}$$

$$= \frac{3x+1}{3x} \times \frac{x^2}{x^2-1}$$

$$= \frac{x(3x+1)}{3(x-1)(x+1)}$$

Guided Practice

Simplify

a $\frac{1+\frac{1}{x}}{1-\frac{1}{x}}$

$\frac{x+1}{x-1}$

b $\frac{2+\frac{1}{x+1}}{3-\frac{1}{x+1}}$

$\frac{2x+3}{3x+2}$

c $\frac{1-\frac{1}{5x}}{1-\frac{1}{x^2}}$

$\frac{x(5x-1)}{5(x-1)(x+1)}$

d $\frac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}-\frac{1}{y}}$

$\frac{x+y}{y-x}$

1 Simplify

a $\dfrac{\frac{1}{x}}{\frac{2}{x}}$

$\frac{1}{2}$

b $\dfrac{\frac{2}{x}}{\frac{1}{x}}$

2

c $\dfrac{\frac{1}{x}}{\frac{1}{x^2}}$

x

d $\dfrac{\frac{1}{x^2}}{\frac{1}{x}}$

$\frac{1}{x}$

e $\dfrac{\frac{x}{y}}{\frac{x}{y^2}}$

y

f $\dfrac{\frac{x^2}{y}}{\frac{x}{y^2}}$

xy

2 Simplify

a $\dfrac{1+\frac{1}{x}}{2}$

$\frac{x+1}{2x}$

b $\dfrac{1+\frac{2}{x}}{2}$

$\frac{x+2}{2x}$

c $\dfrac{1+\frac{1}{x}}{1+\frac{2}{x}}$

$\frac{x+1}{x+2}$

d $\dfrac{1-\frac{1}{x}}{1+\frac{1}{x}}$

$\frac{x-1}{x+1}$

e $\dfrac{1-\frac{1}{x}}{1-\frac{2}{x}}$

$\frac{x-1}{x-2}$

f $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x}}$

$\frac{x+1}{x-1}$

g $\dfrac{2-\frac{1}{x}}{3+\frac{1}{x}}$

$\frac{2x-1}{3x+1}$

h $\dfrac{1-\frac{a}{x}}{1+\frac{a}{x}}$

$\frac{x-a}{x+a}$

3 Explain why $\dfrac{\frac{a}{b}}{\frac{c}{d}} = \dfrac{ad}{bc}$.

Dividing by a fraction is equivalent to multiplying by its reciprocal:

$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$

4 Simplify

a $\dfrac{\frac{1}{x+1}+\frac{1}{x-1}}{\frac{1}{x+1}-\frac{1}{x-1}}$

$-x$

b $\dfrac{\frac{1}{x+2}+\frac{1}{x-2}}{\frac{2}{x^2-4}}$

x

c $\dfrac{1-\frac{1}{x^2}}{1+\frac{1}{x}}$

$\frac{x-1}{x}$

d $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x^2}}$

$\frac{x}{x-1}$

5 Solve for x

$$\frac{1+\frac{1}{x}}{1-\frac{1}{x}}=3$$

$\frac{x+1}{x-1}=3$

$x+1=3x-3$

$4=2x$

$x=2.$

6 Show that the average of $\frac{1}{n}$ and $\frac{1}{n+1}$ can be written as $\frac{2n+1}{2n(n+1)}$.

$\frac{\frac{1}{n}+\frac{1}{n+1}}{2}=\frac{\frac{n+1+n}{n(n+1)}}{2}$

$=\frac{2n+1}{n(n+1)}\times\frac{1}{2}$

$=\frac{2n+1}{2n(n+1)}$

7 In optics, the thin lens formula is $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$.

Show that $f=\frac{uv}{u+v}$, then write this as a compound fraction.

$\frac{1}{f}=\frac{u+v}{uv}$, so $f=\frac{uv}{u+v}$.

This is also the reciprocal of the original $\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$: $f=\frac{1}{\frac{1}{u}+\frac{1}{v}}$

8 Two resistors with resistances $\frac{1}{r}$ and $\frac{1}{s}$ ohms are connected in parallel.

The combined resistance is given by $\frac{1}{\frac{1}{r}+\frac{1}{s}}$. Simplify this expression.

$\frac{1}{\frac{1}{r}+\frac{1}{s}} = \frac{1}{\frac{r+s}{rs}} = \frac{rs}{r+s}$

9 A cyclist travels at $\frac{1}{a}$ of their maximum speed for the first half of a journey and $\frac{1}{b}$ of their maximum speed for the second half. Their average speed as a fraction of maximum speed is $\frac{1}{\frac{1}{2}\left(\frac{1}{a}+\frac{1}{b}\right)}$. Simplify this expression.

$\frac{1}{\frac{1}{2}\cdot\frac{a+b}{ab}} = \frac{1}{\frac{a+b}{2ab}} = \frac{2ab}{a+b}$

10 Simplify

a $\frac{1-\frac{1}{x+1}}{\frac{1}{x}+\frac{1}{x+1}}$

$\frac{x^2}{2x+1}$

b $\frac{\frac{1}{x}+\frac{1}{x+1}}{\frac{1}{x}-\frac{1}{x+1}}$

$2x+1$

c $\frac{\frac{2}{x}+\frac{1}{x+3}}{\frac{3}{x}-\frac{1}{x+3}}$

$\frac{3(x+2)}{2x+9}$

d $\frac{\frac{3}{x+2}-\frac{2}{x+1}}{\frac{5}{x+2}-\frac{4}{x+1}}$

$\frac{x-1}{x-3}$

11 Write as a single fraction, without negative indices, and simplify.

a $a^{-1} - b^{-1}$

$\frac{b-a}{ab}$

b $\frac{1-y^{-1}}{1-y^{-2}}$

$\frac{y}{y+1}$

c $(x^{-2} - y^{-2})^{-1}$

$\frac{x^2y^2}{y^2-x^2}$

d $\frac{a^{-1}+b^{-1}}{a^{-2}-b^{-2}}$

$\frac{ab}{b-a}$

e $x^{-2}y^{-2}(x^2y^{-1} - y^2x^{-1})$

$\frac{x^3-y^3}{x^3y^3}$

f $\frac{(a^2-1)^{-1}}{(a-1)^{-1}}$

$\frac{1}{a+1}$

12 If $x = \frac{1}{t}$, $y = \frac{1}{1-x}$, $z = \frac{y}{y-1}$, show that $z = t$.

$y = \frac{1}{1-x}$

$y = \frac{1}{1-\frac{1}{t}}$

$y = \frac{t}{t-1}$

$z = \frac{y}{y-1}$

$z = \frac{\frac{t}{t-1}}{\frac{t}{t-1}-1}$

$z = \frac{\frac{t}{t-1}}{\frac{t-(t-1)}{t-1}}$

$z = \frac{\frac{t}{t-1}}{\frac{1}{t-1}}$

$z = t$

Challenge Exercise

1 Consider this pattern:

$$\frac{1}{2}-\frac{1}{3}=\frac{1}{6}, \quad \frac{1}{3}-\frac{1}{4}=\frac{1}{12}, \quad \frac{1}{4}-\frac{1}{5}=\frac{1}{20}$$

Prove that, for *any* two consecutive integers, the difference of their reciprocals is the reciprocal of their product.

Let the two consecutive integers be n and $n+1$

$$\frac{1}{n}-\frac{1}{n+1}=\frac{n+1-n}{n(n+1)}=\frac{1}{n(n+1)}$$

2 Consider the pattern:

$$\left(\frac{3}{2}+\frac{2}{3}\right)^2-\left(\frac{3}{2}-\frac{2}{3}\right)^2=4, \quad \left(\frac{7}{6}+\frac{6}{7}\right)^2-\left(\frac{7}{6}-\frac{6}{7}\right)^2=4,$$

Write a rule for this pattern and prove it algebraically.

$$\left(\frac{n+1}{n}+\frac{n}{n+1}\right)^2-\left(\frac{n+1}{n}-\frac{n}{n+1}\right)^2$$

$$=\left(\frac{n^2+2n+1+n^2}{n(n+1)}\right)^2-\left(\frac{n^2+2n+1-n^2}{n(n+1)}\right)^2$$

$$=\left(\frac{2n^2+2n+1}{n(n+1)}\right)^2-\left(\frac{2n+1}{n(n+1)}\right)^2$$

Using difference of two squares,

$$=\left(\frac{2n^2+2n+1}{n(n+1)}+\frac{2n+1}{n(n+1)}\right)\left(\frac{2n^2+2n+1}{n(n+1)}-\frac{2n+1}{n(n+1)}\right)$$

$$=\left(\frac{2n^2+4n+2}{n(n+1)}\right)\left(\frac{2n^2}{n(n+1)}\right)$$

$$=\left(\frac{2(n+1)^2}{n(n+1)}\right)\left(\frac{2n^2}{n(n+1)}\right)$$

By cancelling common factors,

$$=4$$

3 If $x + y = 9$ and $xy = 5$, find the value of:

$$\frac{1}{x} + \frac{1}{y}$$

$\frac{1}{x} + \frac{1}{y}$

$= \frac{x + y}{xy}$

$= \frac{9}{5}$

4 If y is a positive real number and $\left(y + \frac{1}{y}\right)^2 = 5$, find the value of:

a $y + \frac{1}{y}$

y is positive, so $y + \frac{1}{y} = \sqrt{5}$

b $y^2 + \frac{1}{y^2}$

$\left(y + \frac{1}{y}\right)^2 = y^2 + \frac{1}{y^2} + 2 = 5$

$y^2 + \frac{1}{y^2} = 3.$

5 One way to find Pythagorean triads is to do the following:

$\frac{1}{1} + \frac{1}{3} = \frac{4}{3} \quad \rightarrow 4^2 + 3^2 = 5^2$

$\frac{1}{3} + \frac{1}{5} = \frac{8}{5} \quad \rightarrow 8^2 + 15^2 = 17^2$

$\frac{1}{5} + \frac{1}{7} = \frac{12}{35} \quad \rightarrow 12^2 + 35^2 = 37^2$

Prove this will always generate a Pythagorean triad.

Generalisation: $\frac{1}{2n-1} + \frac{1}{2n+1} = \frac{4n}{4n^2 - 1}$

Proof: LHS $= \frac{(2n+1) + (2n-1)}{(2n-1)(2n+1)} = \frac{4n}{4n^2 - 1} =$ RHS

For the Pythagorean triples: $(4n)^2 + (4n^2 - 1)^2 = (4n^2 + 1)^2$.

Proof: LHS $= 16n^2 + 16n^4 - 8n^2 + 1 = (4n^2 + 1)^2 =$ RHS.

6 Given $\frac{p}{q} = \frac{2}{3}$, prove that:

a

$$\frac{p-q}{q} = -\frac{1}{3}$$

$LHS = \frac{p-q}{q}$

$= \frac{p}{q} - 1$

$= \frac{2}{3} - 1$

$= -\frac{1}{3}$

b

$$\frac{p-q}{p+q} = -\frac{1}{5}$$

$LHS = \frac{p-q}{p+q}$

$= \frac{\frac{p}{q} - 1}{\frac{p}{q} + 1}$

$= \frac{\frac{2}{3} - 1}{\frac{2}{3} + 1}$

$= -\frac{1}{5}$

c

$$\frac{p^2+q^2}{p^2-q^2} = -\frac{13}{5}$$

$LHS = \frac{p^2+q^2}{p^2-q^2}$

$= \frac{\frac{p^2}{q^2} + 1}{\frac{p^2}{q^2} - 1}$

$= \frac{\frac{4}{9} + 1}{\frac{4}{9} - 1}$

$= -\frac{13}{5}$

7 Given four numbers, a, b, c, d satisfy the proportion $\frac{a}{b} = \frac{c}{d}$, prove they also satisfy:

From $\frac{a}{b} = \frac{c}{d}$, $a = \frac{bc}{d}$ (so $ad = bc$).

a

$$\frac{a}{c} = \frac{b}{d}$$

$LHS = \frac{a}{c}$

$= \frac{bc/d}{c}$

$= \frac{b}{d}$

b

$$\frac{a+b}{b} = \frac{c+d}{d}$$

$LHS = \frac{a+b}{b}$

$= \frac{a}{b} + 1$

$= \frac{c}{d} + 1$

$= \frac{c+d}{d}$

c

$$\frac{a-b}{b} = \frac{c-d}{d}$$

$LHS = \frac{a-b}{b}$

$= \frac{a}{b} - 1$

$= \frac{c}{d} - 1$

$= \frac{c-d}{d}$

d

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$LHS = \frac{a+b}{a-b}$

$= \frac{(a+b)/b}{(a-b)/b}$

$= \frac{(c+d)/d}{(c-d)/d}$

$= \frac{c+d}{c-d}$

e

$$\frac{a}{b} = \frac{a+c}{b+d}$$

$a + c = \frac{bc}{d} + c$

$= \frac{c(b+d)}{d}$

$\therefore RHS = \frac{a+c}{b+d} = \frac{c}{d} = \frac{a}{b}$

13 Simplify

$$\left(2-\frac{3n}{m}+\frac{9n^2-2m^2}{m^2+2mn}\right)\div\left(\frac{1}{m}-\frac{1}{m-2n-\frac{4n^2}{m+n}}\right)$$

$$=\left(\frac{2m(m+2n)-3n(m+2n)+9n^2-2m^2}{m(m+2n)}\right)\div\left(\frac{1}{m}-\frac{1}{\frac{(m-2n)(m+n)-4n^2}{m+n}}\right)$$

$$=\left(\frac{2m^2+4mn-3mn-6n^2+9n^2-2m^2}{m(m+2n)}\right)\div\left(\frac{1}{m}-\frac{m+n}{m^2-mn-6n^2}\right)$$

$$=\frac{n(m+3n)}{m(m+2n)}\div\left(\frac{1}{m}-\frac{m+n}{(m-3n)(m+2n)}\right)$$

$$=\frac{n(m+3n)}{m(m+2n)}\div\left(\frac{(m-3n)(m+2n)-m(m+n)}{m(m-3n)(m+2n)}\right)$$

$$=\frac{n(m+3n)}{m(m+2n)}\div\left(\frac{m^2-mn-6n^2-m^2-mn}{m(m-3n)(m+2n)}\right)$$

$$=\frac{n(m+3n)}{m(m+2n)}\div\frac{-2n(m+3n)}{m(m-3n)(m+2n)}$$

$$=\frac{n(m+3n)}{m(m+2n)}\times\frac{m(m-3n)(m+2n)}{-2n(m+3n)}$$

$$=\frac{m-3n}{-2}$$

$$=\frac{3n-m}{2}$$

14 Simplify

$$\frac{1}{x+\frac{1}{x+2}} \times \frac{1}{x+\frac{1}{x-2}} \div \frac{x-\frac{4}{x}}{x^2-2+\frac{1}{x^2}}$$

$$\frac{1}{x+\frac{1}{x+2}} \times \frac{1}{x+\frac{1}{x-2}} \div \frac{x-\frac{4}{x}}{x^2-2+\frac{1}{x^2}}$$

$$= \frac{1}{\frac{x(x+2)+1}{x+2}} \times \frac{1}{\frac{x(x-2)+1}{x-2}} \div \frac{\frac{x^2-4}{x}}{\frac{x^4-2x^2+1}{x^2}}$$

$$= \frac{x+2}{x^2+2x+1} \times \frac{x-2}{x^2-2x+1} \div \frac{\frac{(x-2)(x+2)}{x}}{\frac{(x^2-1)^2}{x^2}}$$

$$= \frac{x+2}{(x+1)^2} \times \frac{x-2}{(x-1)^2} \div \left(\frac{(x-2)(x+2)}{x} \cdot \frac{x^2}{(x-1)^2(x+1)^2}\right)$$

$$= \frac{(x+2)(x-2)}{(x+1)^2(x-1)^2} \div \frac{x(x-2)(x+2)}{(x-1)^2(x+1)^2}$$

$$= \frac{(x+2)(x-2)}{(x+1)^2(x-1)^2} \times \frac{(x-1)^2(x+1)^2}{x(x-2)(x+2)}$$

$$= \frac{1}{x}$$